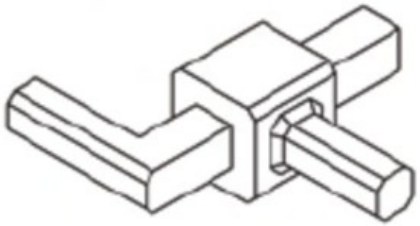
Manipulators and robots. Direct kinematics. Inverse kinematics

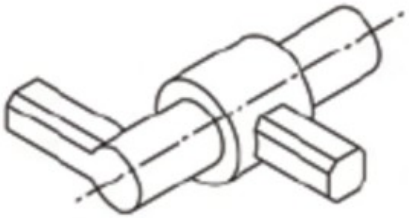
Basic Robot Components

- Links – rigid or flexible
- Joints – different kinematic types
- Actuators – rotational or translational
- Sensors – motion, force, vision, etc.
- End-Effector
- Computer
- Software
- Human-machine interfaces
- Power source

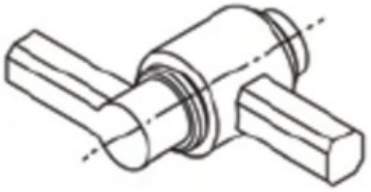
Kinematic pair is shown consisting of two consecutive segments of a robot mechanism



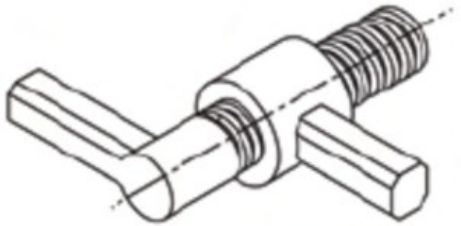
Prismatic (P)



Cylindrical (C)



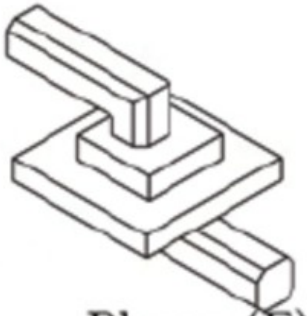
Revolute (R)



Helical (H)



Spherical (S)

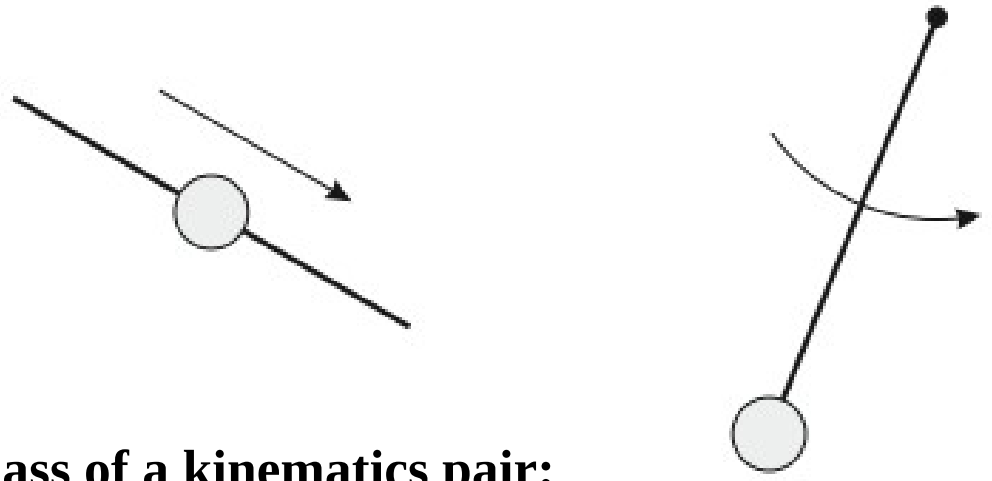


Planar (E)

Simple movements:

Particle moving along a line

Pendulum with a rigid segment, which is swinging in a plane



Class of a kinematics pair:

V – translation or rotation - just ONE movement

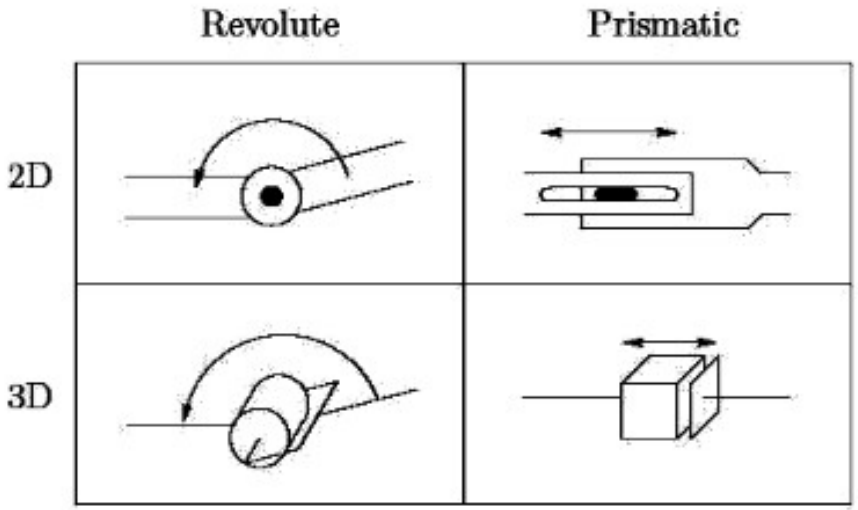
IV – 2 possible movements

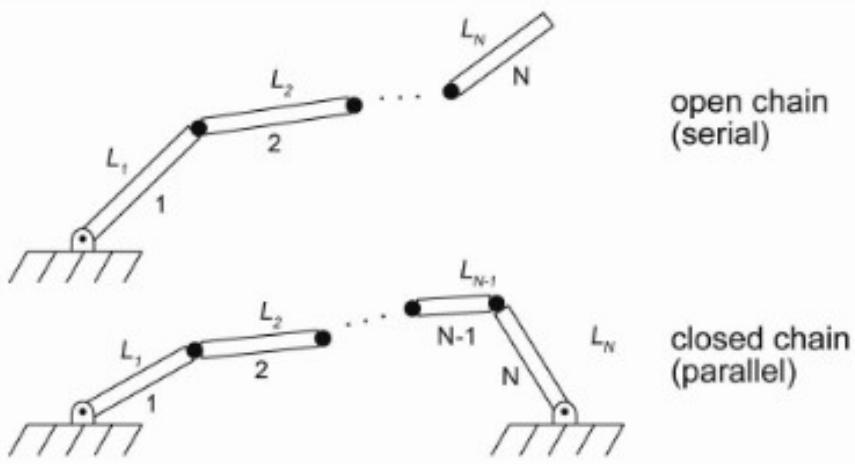
III – 3

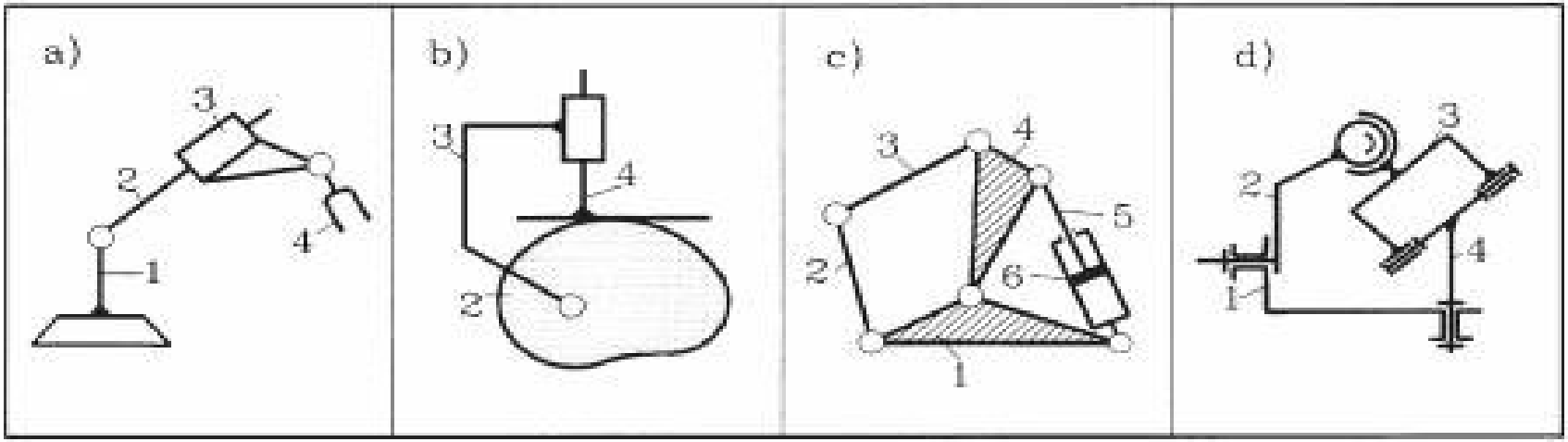
II – 4

I – 5

Open and closed robot chains (mechanisms):







Degree of freedom ***DOF*** or degree of movement ***W***

In mechanics, degrees of freedom (DOF) is the number of independent variables that define the possible positions or motions of a mechanical system in space. DOF measurements assume that the mechanism is both rigid and unconstrained, whether it operates in two-dimensional or three-dimensional space. The number of degrees of freedom is equal to the total number of independent displacements or aspects of motion.

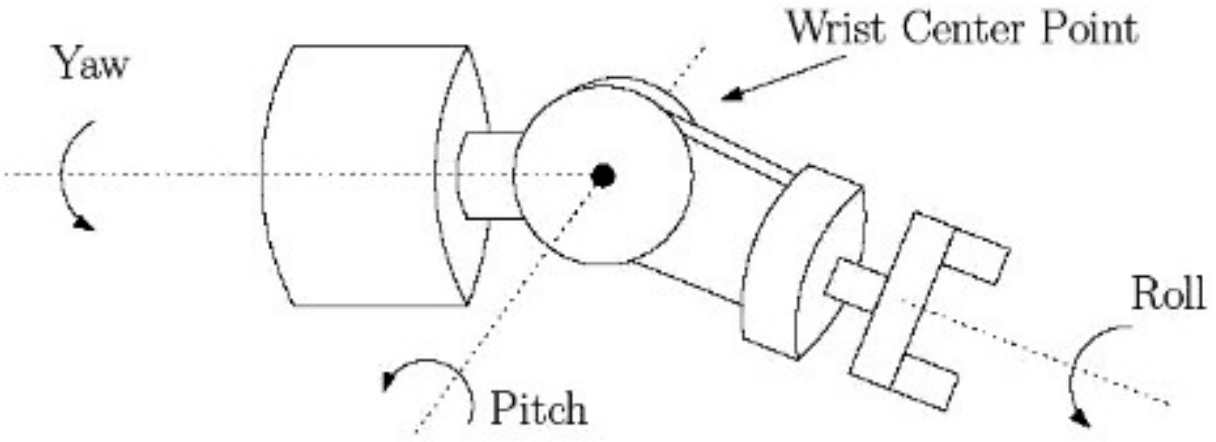
$$W = 6(n - 1) - 5p_5 - 4p_4 - 3p_3 - 2p_2 - p_1$$

where:

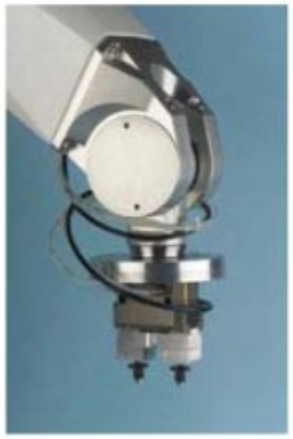
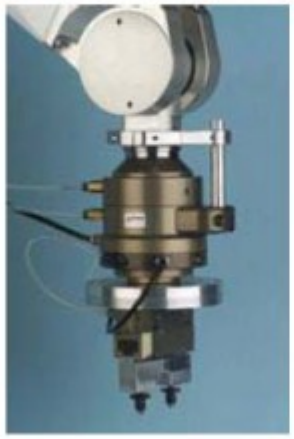
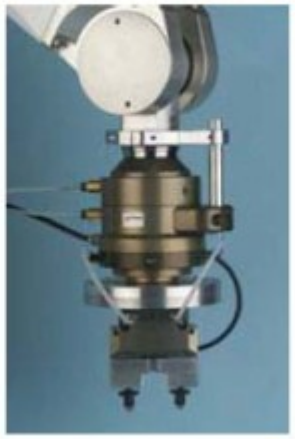
n – number of the moving arms;

p_i – number of pairs i -th class

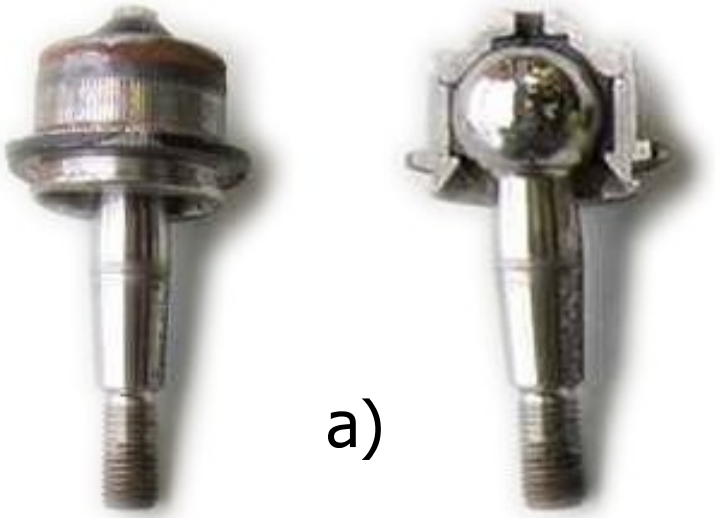
WRIST: concentration of links and joints forming the 'arm' with multiple DOFs



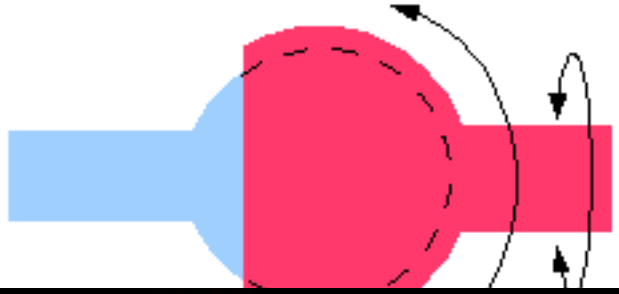
Industrial Robot Wrist

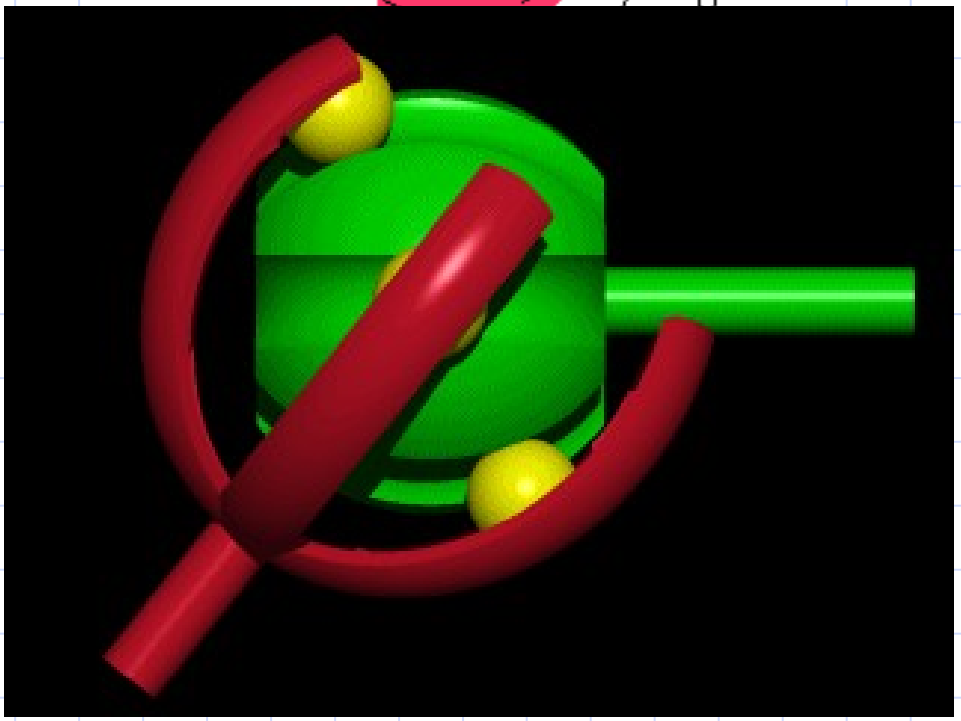

What we see?

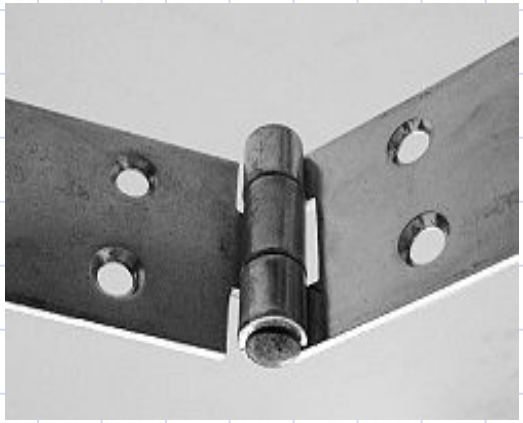


a)



b)





c)

d)

What about this?

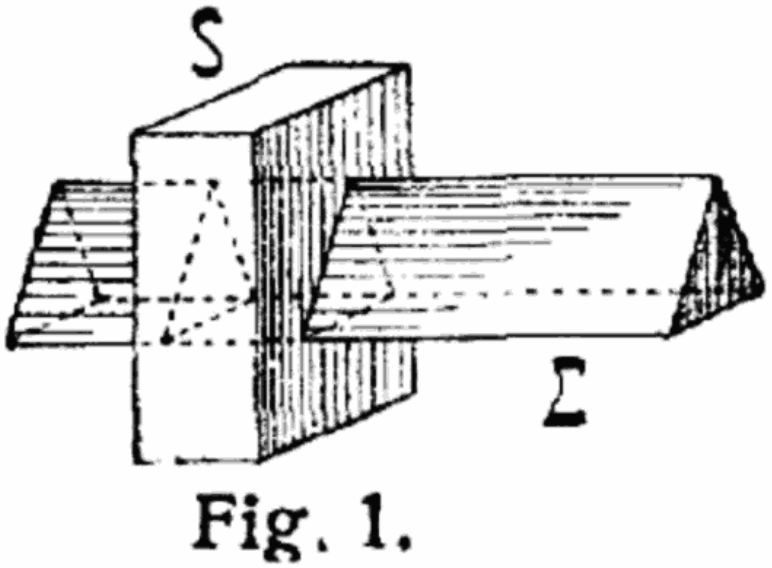
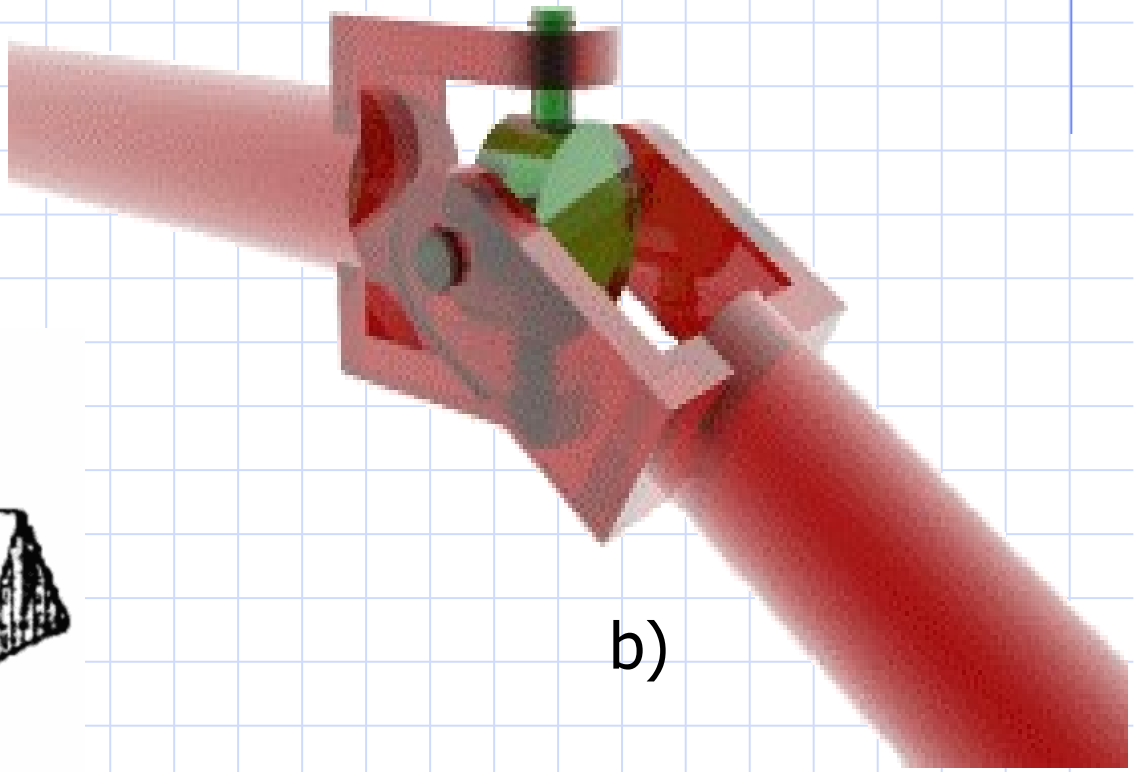


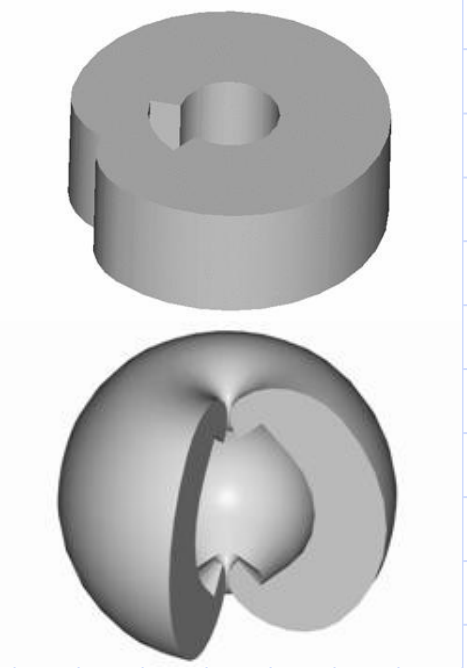
Fig. 1.

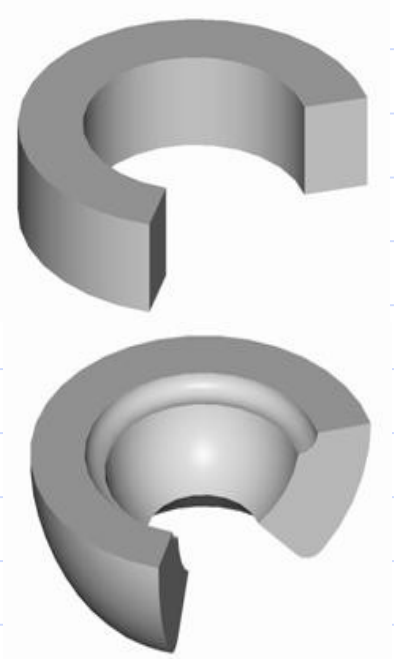
a)



b)

Most popular robotics ork spaces



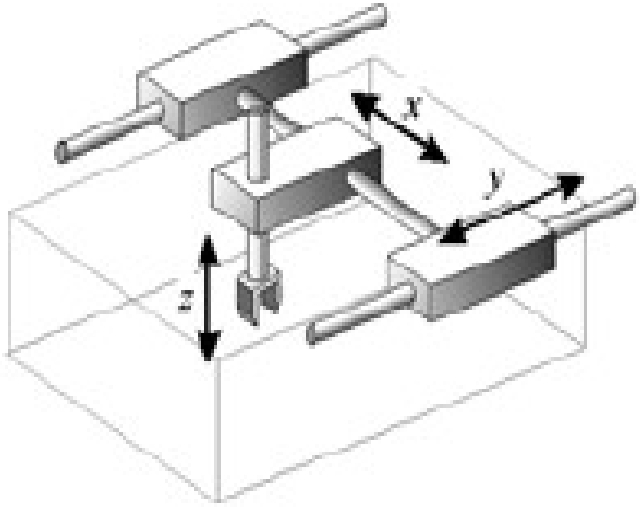


Definitions

- End-effector/Tool
  - Device that is in direct contact with the environment. Usually very task-specific
- Configuration
  - Complete specification of every point on a manipulator
  - set of all possible configurations is the *configuration space*
  - For rigid links, it is sufficient to specify the configuration space by the joint angles $q = [q_1 \quad q_2 \quad \dots \quad q_n]^T$
- State space
  - Current configuration (joint positions q) and velocities
- Work space
  - The reachable space the tool can achieve
    - Reachable workspace
    - Dextrous workspace

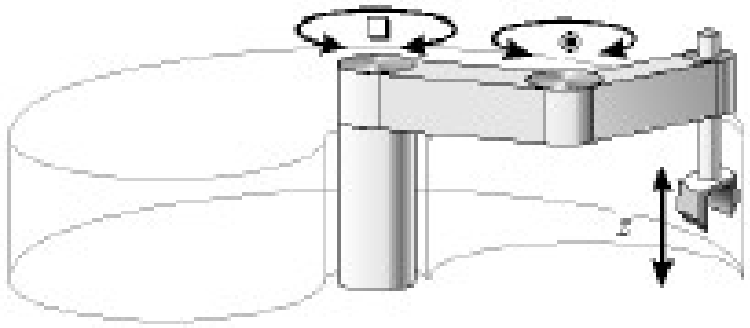
TTT configuration





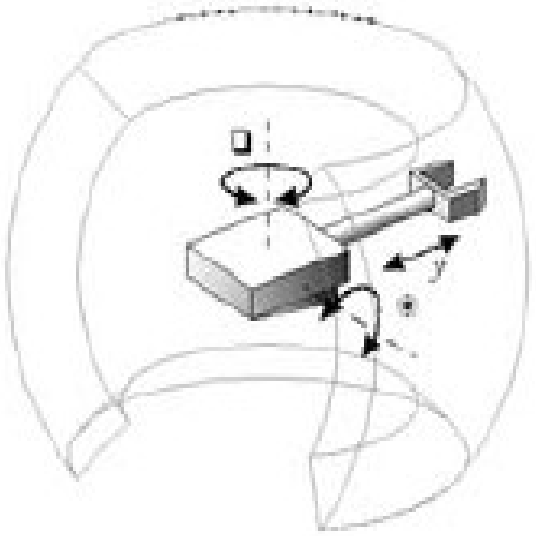
SCARA RRT





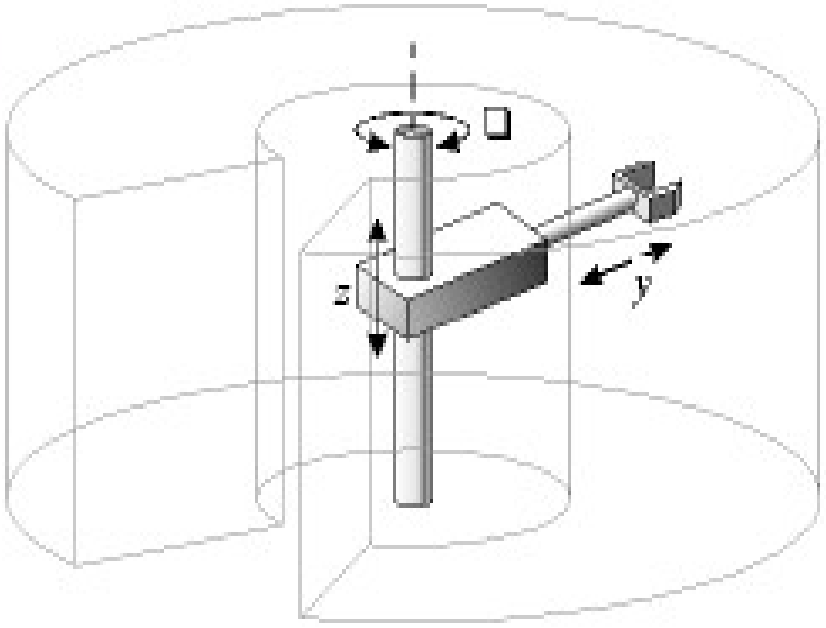
RRT (sferic)





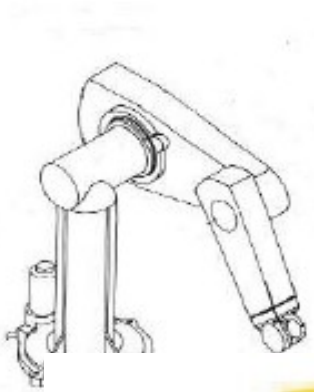
RTT (cylindric)



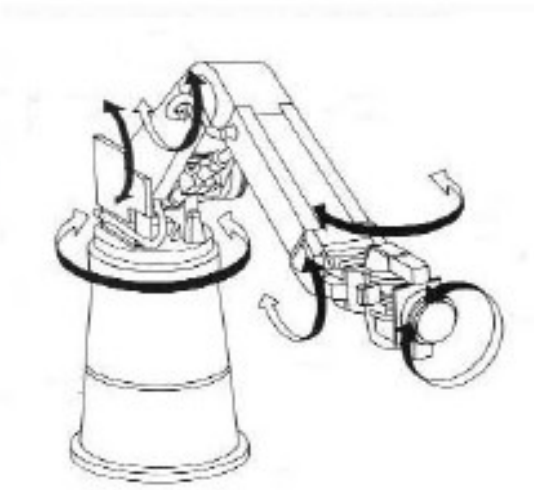


RRR (arm)

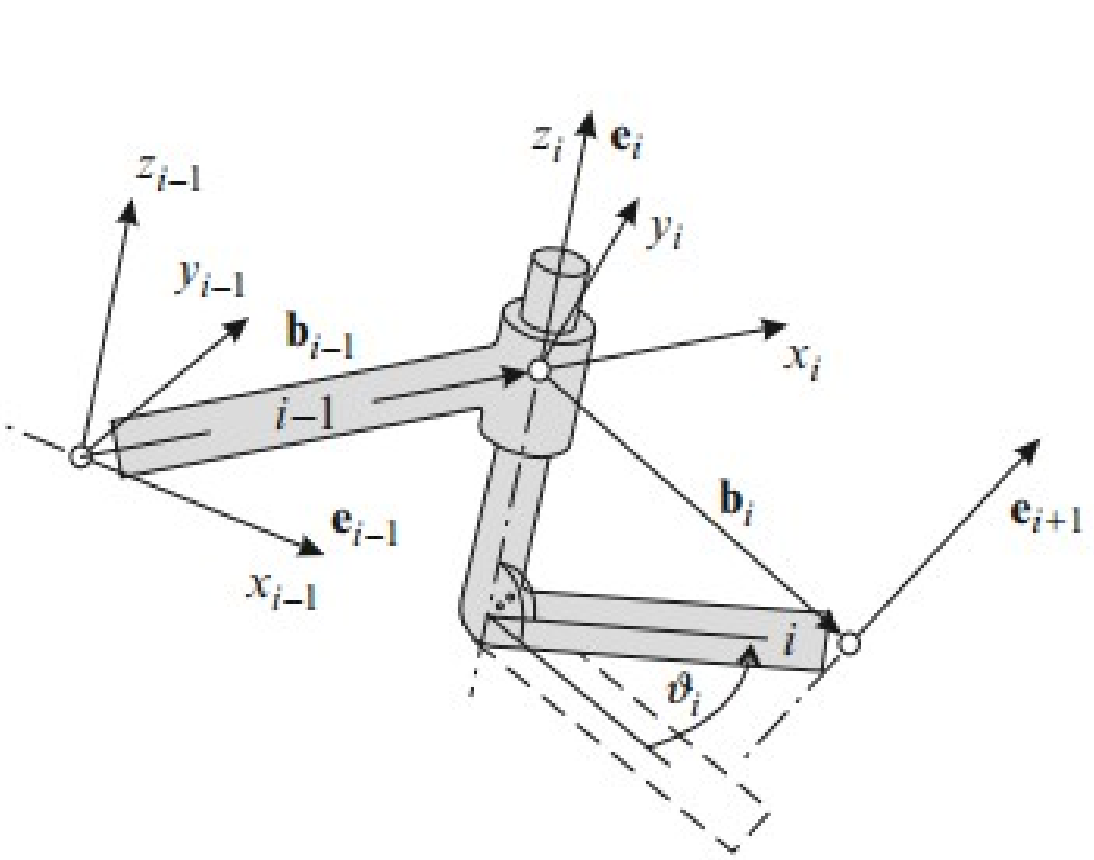








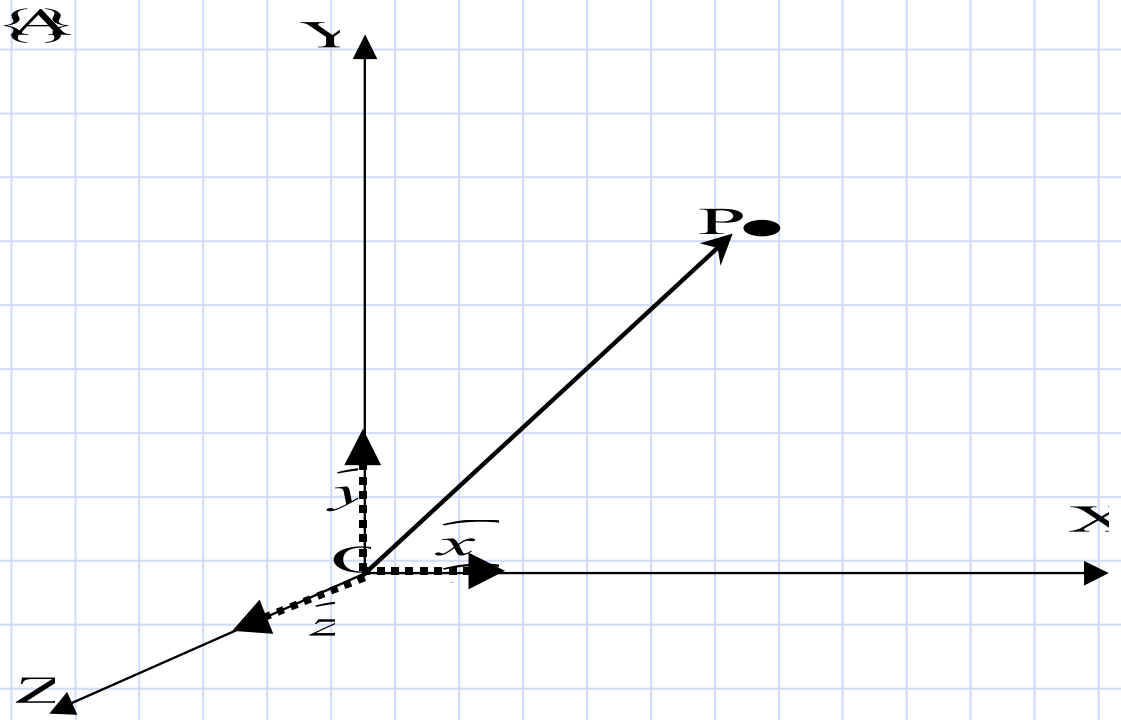
Direct and Invers Kinematics



Direct kinematics in the case of a two-segment robot represents the calculation of the position of the robot end-point from the known joint angles. **Inverse kinematics** calculates the joint variables from the known position of the robot end-point.

Position

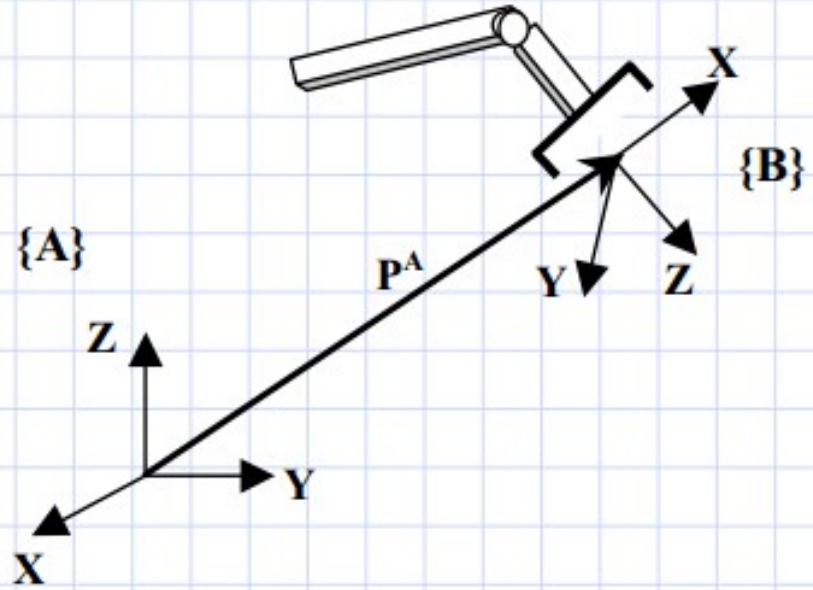
$$ {}^{A}P = \begin{bmatrix} P_x \\ P_y \\ P_z \end{bmatrix} \quad \{A\} $$



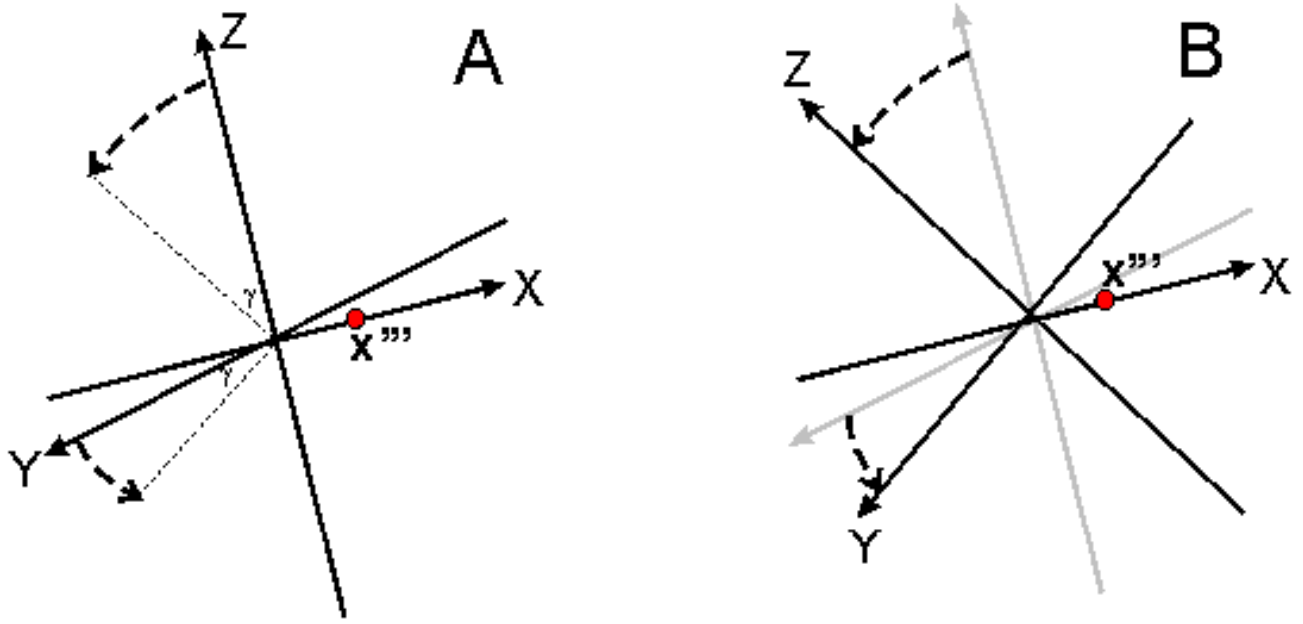
Orientation

Macierz obrotu

$$ {}^{A}_{B}R = \left[{}^{A}\hat{X}_B \; {}^{A}\hat{Y}_B \; {}^{A}\hat{Z}_B \right] = \begin{bmatrix} r_{11} r_{12} r_{13} \\ r_{21} r_{22} r_{23} \\ r_{31} r_{32} r_{33} \end{bmatrix} $$



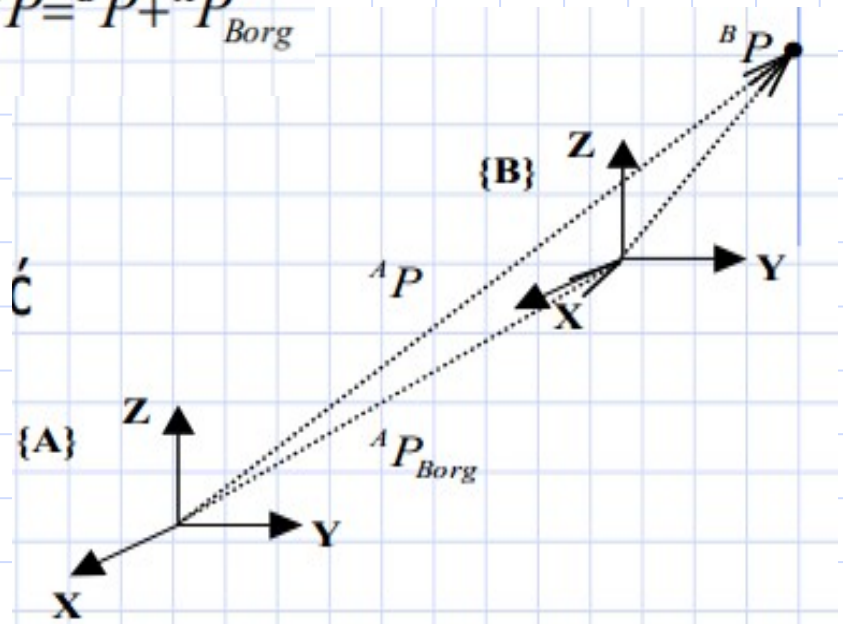
$$ {}^{A}_{B}R = \begin{bmatrix} {}^{A}X_B & {}^{A}Y_B & {}^{A}Z_B \end{bmatrix} = \begin{bmatrix} x_B x_A & y_B x_A & z_B x_A \\ x_B y_A & y_B y_A & z_B y_A \\ x_B z_A & y_B z_A & z_B z_A \end{bmatrix} $$



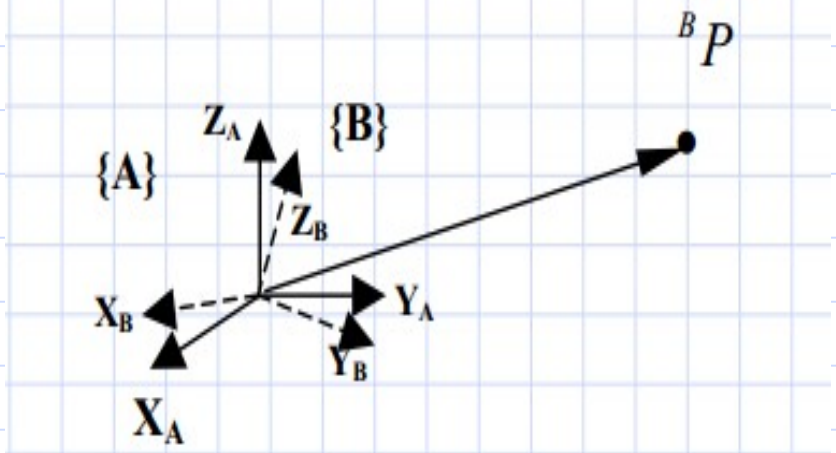
$$ {}^{B}_{A}R = \begin{bmatrix} {}^{B}X_A & {}^{B}Y_A & {}^{B}Z_A \end{bmatrix} = ¿ \begin{bmatrix} {}^{A}X_{B^T} \end{bmatrix} \begin{bmatrix} {}^{A}Y_{B^T} \end{bmatrix} $$

$$ {}^{A}_{B}R = \begin{bmatrix} {}^{A}X_B & {}^{A}Y_B & {}^{A}Z_B \end{bmatrix} = \begin{bmatrix} x_B x_A & y_B x_A & z_B x_A \\ x_B y_A & y_B y_A & z_B y_A \\ x_B z_A & y_B z_A & z_B z_A \end{bmatrix} $$

$${}^{A}P = {}^{B}P + {}^{A}P_{Borg}$$



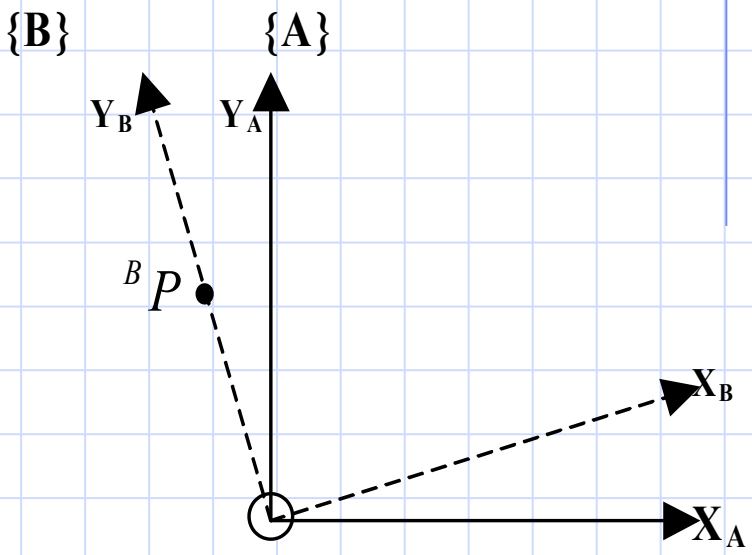
$${}^{A}P = {}^{A}_{B}R \cdot {}^{B}P$$



Task 1

{B} was rotated 30` R(Z_A).

Find ${}^{A}P$, if ${}^{B}P$ lays on Y_B and is long 7.



Task 2

{B} was rotated 90` R (Y_A). Find ${}^{A}P$, if ${}^{B}P$ is $[-3; 5; -8]^T$.

Homogenous transformation matrice T^B

A unified mathematical description of translational and rotational displacements. The translational displacement d, given by the vector $d = ai + bj + ck$, can be described also by the following homogenous transformation matrix ***H***

$$\mathbf{H} = Trans(a,b,c) = \begin{bmatrix} 1 & 0 & 0 & a \\ 0 & 1 & 0 & b \\ 0 & 0 & 1 & c \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

When using homogenous transformation matrices an arbitrary vector has the following 4× 1 form: $q = [x\ y\ z\ 1]^T$ A translational displacement of vector ***q*** for a distance ***d*** is obtained by multiplying the vector ***q*** with the matrix ***H***

The translation, which is presented by multiplication with a homogenous matrics:

$$\mathbf{v} = \begin{bmatrix} 1 & 0 & 0 & a \\ 0 & 1 & 0 & b \\ 0 & 0 & 1 & c \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} x+a \\ y+b \\ z+c \\ 1 \end{bmatrix}$$

$$Trans_{x,a} = \begin{pmatrix} 1 & 0 & 0 & a \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$Rot_{x,\theta} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos(\theta) & -\sin(\theta) & 0 \\ 0 & \sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$Trans_{y,a} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & a \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$Rot_{y,\theta} = \begin{pmatrix} \cos(\theta) & 0 & \sin(\theta) & 0 \\ 0 & 1 & 0 & 0 \\ -\sin(\theta) & 0 & \cos(\theta) & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$Trans_{z,a} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & a \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$Rot_{z,\theta} = \begin{pmatrix} \cos(\theta) & -\sin(\theta) & 0 & 0 \\ \sin(\theta) & \cos(\theta) & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

When we have a movement sequence:

$$
\begin{aligned}
R_2^0 &= R_1^0 R_2^1 \\
l_2^0 &= R_1^0 l_2^1 + l_1^0
\end{aligned}
$$

$$
\begin{pmatrix} R_1^0 & l_1^0 \\ 0 & 1 \end{pmatrix}
\begin{pmatrix} R_2^1 & l_2^1 \\ 0 & 1 \end{pmatrix}
=
\begin{pmatrix} R_1^0 R_2^1 & R_1^0 l_2^1 + l_1^0 \\ 0 & 1 \end{pmatrix}
$$

$$
T_b^a = T_c^a T_b^c
$$

$$
(T_b^a)^{-1} = T_a^b = \begin{pmatrix} (R_b^a)^T & -(R_b^a)^T l_b^a \\ 0 & 1 \end{pmatrix}
$$

Task 3

{B} was rotated R(Z_A) = 30’ and translated: -2 in direction X_A and -8 in Y_A. Find ${}^{A}P$, if ${}^{B}P$ is: $[-1; -39; 5]^T$.

Task 4

{B} was rotated R(X_A) = 45’ and translated: 5 in direction Y_A and -8 in Z_A. Find ${}^{A}P$, if ${}^{B}P$ is: $[4; -2; 17]^T$.